

Step 1: Problem Framing & Theory Notes

1. What is Regression? Difference from Classification

- Regression is a supervised learning technique used to predict continuous numerical values such as house prices or salary.
 - Classification is used to predict categorical outcomes like Yes/No, Spam/Not Spam.
 - Example (India): Predicting EMI amount for a home loan in HDFC (Regression).
 - Example (India): Predicting whether a loan will be approved or not (Classification).
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2. Difference between Linear, Multiple Linear, and Regularised Regression

- Simple Linear Regression uses one independent variable to predict the target.
 - Multiple Linear Regression uses multiple independent variables for prediction.
 - Ridge Regression (L2) reduces model complexity by shrinking coefficients.
 - Lasso Regression (L1) can shrink some coefficients to zero, helping in feature selection.
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3. Overfitting vs Underfitting

- Overfitting occurs when the model learns noise from training data and performs poorly on new data.
 - Underfitting occurs when the model is too simple and cannot capture patterns in the data.
 - Regularisation (L1/L2) helps by penalizing large coefficients and reducing complexity.
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4. RMSE, MAE, and R^2

- RMSE (Root Mean Squared Error) measures the square root of average squared errors and penalizes large errors more.
- MAE (Mean Absolute Error) measures average absolute differences between predicted and actual values.

- R^2 (R-squared) shows how well the model explains variance in the data.
 - RMSE is preferred when large errors are more important to penalize.
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5. Cross-Validation (k-Fold)

- Cross-validation splits data into multiple parts and trains/tests the model multiple times.
- In k-Fold, data is divided into k subsets and each subset is used once as test data.
- It provides more reliable performance compared to a single train-test split.
- It reduces variance and helps detect overfitting.